

Lebsheet-4

1) Create a class Student with instance fields name, age and a class variable count to track number of students. Use local variables to assign values and display details.

```
class Student {
    // Instance fields
    String name;
    int age;

    // Class variable
    static int count = 0;

    // Constructor
    Student(String name, int age) {
        // Local variables
        String studentName = name;
        int studentAge = age;

        this.name = studentName;
        this.age = studentAge;

        count++;
    }

    // Display student details
    void displayDetails() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
    }

    public static void main(String[] args) {
        Student s1 = new Student("Rahul", 20);
        Student s2 = new Student("Priya", 21);

        s1.displayDetails();
        s2.displayDetails();

        System.out.println("Total number of students: " + count);
    }
}
```

Output-

Name: Arpit

Age: 20

Name: Priya

Age: 21

Total number of students: 2

2) Create a class Rectangle with instance fields length, breadth, a class variable shapeName = "Rectangle". Write methods with local variables to calculate and display the area.

```
class Rectangle {
    // Instance fields
    double length;
    double breadth;

    // Class variable
    static String shapeName = "Rectangle";

    // Constructor
    Rectangle(double length, double breadth) {
        this.length = length;
        this.breadth = breadth;
    }

    // Method to calculate and display area
    void displayArea() {
        // Local variable
        double area = length * breadth;

        System.out.println("Shape: " + shapeName);
        System.out.println("Length: " + length);
        System.out.println("Breadth: " + breadth);
        System.out.println("Area: " + area);
    }

    public static void main(String[] args) {
        Rectangle r = new Rectangle(10, 5);

        r.displayArea();
    }
}
```

Output-
Shape: Rectangle

Length: 10.0

Breadth: 5.0

Area: 50.0

3) Create a class Circle with instance field radius, a class variable PI = 3.14. Use local variables to calculate area and circumference.

```
class Circle {
    // Instance field
    double radius;

    // Class variable
    static double PI = 3.14;

    // Constructor
    Circle(double radius) {
        this.radius = radius;
    }

    // Method to calculate area and circumference
    void calculate() {
        // Local variables
        double area = PI * radius * radius;
        double circumference = 2 * PI * radius;

        System.out.println("Radius: " + radius);
        System.out.println("Area: " + area);
        System.out.println("Circumference: " + circumference);
    }

    public static void main(String[] args) {
        Circle c = new Circle(5);

        c.calculate();
    }
}
```

Output-

Radius: 5.0

Area: 78.5

Circumference: 31.400000000000002

4) Create a class Employee with instance fields empld, salary, and a class variable companyName. Display employee details using local variables.

```

class Employee {
    // Instance fields
    int empld;
    double salary;

    // Class variable
    static String companyName = "ABC Company";

    // Constructor
    Employee(int empld, double salary) {
        this.empld = empld;
        this.salary = salary;
    }

    // Method to display employee details using local variables
    void displayDetails() {
        int id = empld;          // Local variable
        double sal = salary;      // Local variable
        String company = companyName; // Local variable

        System.out.println("Employee ID: " + id);
        System.out.println("Salary: " + sal);
        System.out.println("Company Name: " + company);
    }

    public static void main(String[] args) {
        Employee e1 = new Employee(101, 50000);
        Employee e2 = new Employee(102, 60000);

        e1.displayDetails();
        System.out.println();
        e2.displayDetails();
    }
}

```

Output-

```

Employee ID: 101
Salary: 50000.0
Company Name: ABC Company

```

```

Employee ID: 102
Salary: 60000.0
Company Name: ABC Company

```

5) Create a class Book with fields title, author, price, and a class variable publisher. Use local variables to set and display book details.

```
class Book {
    // Instance fields
    String title;
    String author;
    double price;

    // Class variable
    static String publisher = "ABC Publications";

    // Method to set and display book details using local variables
    void setDetails() {
        String localTitle = "Java Programming";
        String localAuthor = "James Gosling";
        double localPrice = 599.00;

        title = localTitle;
        author = localAuthor;
        price = localPrice;
    }

    void displayDetails() {
        System.out.println("Title    : " + title);
        System.out.println("Author   : " + author);
        System.out.println("Price    : ₹" + price);
        System.out.println("Publisher : " + publisher);
    }

    public static void main(String[] args) {
        Book b = new Book();

        b.setDetails();
        b.displayDetails();
    }
}
```

Output-

```
Title    : Java Programming
Author   : James Gosling
Price    : ₹599.0
Publisher : ABC Publications
```

6) Create a class Car with fields brand, mileage, and a class variable wheels = 4. Display details using local variables.

```
class Car {
    // Instance fields
    String brand;
    double mileage;

    // Class variable
    static int wheels = 4;

    // Method to set details using local variables
    void setDetails() {
        String localBrand = "Toyota";
        double localMileage = 18.5;

        brand = localBrand;
        mileage = localMileage;
    }

    // Method to display details
    void displayDetails() {
        System.out.println("Brand   : " + brand);
        System.out.println("Mileage : " + mileage + " km/l");
        System.out.println("Wheels  : " + wheels);
    }

    public static void main(String[] args) {
        Car c = new Car();

        c.setDetails();
        c.displayDetails();
    }
}
```

Output-

```
Brand   : Toyota
Mileage : 18.5 km/l
Wheels  : 4
```

7) Create a class BankAccount with fields accountNumber, balance, and a class variable bankName. Use local variables in deposit method.

```

class BankAccount {
    // Instance fields
    long accountNumber;
    double balance;

    // Class variable
    static String bankName = "State Bank";

    // Deposit method using a local variable
    void deposit() {
        double depositAmount = 5000.00; // Local variable
        balance = balance + depositAmount;
    }

    // Display account details
    void displayDetails() {
        System.out.println("Bank Name    : " + bankName);
        System.out.println("Account Number : " + accountNumber);
        System.out.println("Balance      : ₹" + balance);
    }

    public static void main(String[] args) {
        BankAccount account = new BankAccount();

        account.accountNumber = 1234567890;
        account.balance = 10000.00;

        account.deposit();
        account.displayDetails();
    }
}

```

Output-

```

Bank Name    : State Bank
Account Number : 1234567890
Balance      : ₹15000.0

```

8) Create a class Pen with fields color, type, and a class variable manufacturer. Use local variables to display pen details.

```

class Pen {
    // Instance fields

```

```

String color;
String type;

// Class variable
static String manufacturer = "Cello";

// Method using local variables to display details
void displayDetails() {
    String localColor = "Blue";
    String localType = "Ballpoint";

    color = localColor;
    type = localType;

    System.out.println("Color      : " + color);
    System.out.println("Type      : " + type);
    System.out.println("Manufacturer : " + manufacturer);
}

public static void main(String[] args) {
    Pen p = new Pen();
    p.displayDetails();
}
}

```

Output-

```

Color      : Blue
Type      : Ballpoint
Manufacturer : Cello

```

9) Create a class Laptop with fields brand, RAM, and a class variable os = "Windows". Use local variables to display details.

```

class Laptop {
    String brand;
    int RAM;
    static String os = "Windows";

    void displayDetails() {
        // Local variables
        String laptopBrand = brand;
        int laptopRAM = RAM;
        String laptopOS = os;
    }
}

```

```

        System.out.println("Brand: " + laptopBrand);
        System.out.println("RAM: " + laptopRAM + " GB");
        System.out.println("OS: " + laptopOS);
    }

    public static void main(String[] args) {
        Laptop laptop = new Laptop();

        laptop.brand = "Dell";
        laptop.RAM = 16;

        laptop.displayDetails();
    }
}

```

Output-

```

Brand: Dell
RAM: 16 GB
OS: Windows

```

10) Create a class Movie with fields name, genre, rating, and a class variable industry. Display movie details using local variables.

```

class Movie {
    String name;
    String genre;
    double rating;
    static String industry = "Bollywood";

    void displayDetails() {
        // Local variables
        String movieName = name;
        String movieGenre = genre;
        double movieRating = rating;
        String movieIndustry = industry;

        System.out.println("Movie Name: " + movieName);
        System.out.println("Genre: " + movieGenre);
        System.out.println("Rating: " + movieRating);
        System.out.println("Industry: " + movieIndustry);
    }

    public static void main(String[] args) {

```

```

    Movie movie = new Movie();

    movie.name = "3 Idiots";
    movie.genre = "Comedy Drama";
    movie.rating = 8.4;

    movie.displayDetails();
}
}

```

Output-

Movie Name: 3 Idiots
Genre: Comedy Drama
Rating: 8.4
Industry: Bollywood

11) Create a class Calculator with methods for add, subtract, multiply, divide using local variables. Add a class variable operationsCount to count total operations done.

```

class Calculator {
    static int operationsCount = 0;

    void add(int a, int b) {
        int result = a + b; // local variable
        operationsCount++;
        System.out.println("Addition: " + result);
    }

    void subtract(int a, int b) {
        int result = a - b; // local variable
        operationsCount++;
        System.out.println("Subtraction: " + result);
    }

    void multiply(int a, int b) {
        int result = a * b; // local variable
        operationsCount++;
        System.out.println("Multiplication: " + result);
    }

    void divide(int a, int b) {
        int result = a / b; // local variable
        operationsCount++;
        System.out.println("Division: " + result);
    }
}

```

```

    }

    public static void main(String[] args) {
        Calculator calc = new Calculator();

        calc.add(20, 10);
        calc.subtract(20, 10);
        calc.multiply(20, 10);
        calc.divide(20, 10);

        System.out.println("Total Operations: " + operationsCount);
    }
}

```

Output-

Addition: 30

Subtraction: 10

Multiplication: 200

Division: 2

Total Operations: 4

12) Create a class StudentMarks with fields marks1, marks2, marks3, and a class variable universityName. Use local variables to calculate average.

```

class StudentMarks {
    int marks1;
    int marks2;
    int marks3;

    static String universityName = "ABC University";

    void displayDetails() {
        // Local variables
        int m1 = marks1;
        int m2 = marks2;
        int m3 = marks3;
        double average = (m1 + m2 + m3) / 3.0;

        System.out.println("University: " + universityName);
        System.out.println("Marks 1: " + m1);
        System.out.println("Marks 2: " + m2);
        System.out.println("Marks 3: " + m3);
        System.out.println("Average: " + average);
    }
}

```

```

public static void main(String[] args) {
    StudentMarks student = new StudentMarks();

    student.marks1 = 80;
    student.marks2 = 75;
    student.marks3 = 90;

    student.displayDetails();
}
}

```

Output-

```

University: ABC University
Marks 1: 80
Marks 2: 75
Marks 3: 90
Average: 81.66666666666667

```

13) Create a class Temperature with a field celsius and a class variable scale = "Celsius". Use local variables to convert into Fahrenheit and Kelvin.

```

class Temperature {
    double celsius;
    static String scale = "Celsius";

    void convertTemperature() {
        // Local variables
        double fahrenheit = (celsius * 9 / 5) + 32;
        double kelvin = celsius + 273.15;

        System.out.println("Temperature: " + celsius + " " + scale);
        System.out.println("Fahrenheit: " + fahrenheit);
        System.out.println("Kelvin: " + kelvin);
    }

    public static void main(String[] args) {
        Temperature temp = new Temperature();

        temp.celsius = 25;

        temp.convertTemperature();
    }
}

```

Output-

Temperature: 25.0 Celsius

Fahrenheit: 77.0

Kelvin: 298.15

14) Create a class Person with fields name, age and a class variable country. Use local variables to check voting eligibility.

```
class Person {
    String name;
    int age;

    static String country = "India";

    void checkEligibility() {
        // Local variables
        String personName = name;
        int personAge = age;
        boolean eligible = personAge >= 18;

        System.out.println("Name: " + personName);
        System.out.println("Age: " + personAge);
        System.out.println("Country: " + country);
        System.out.println("Voting Eligible: " + eligible);
    }

    public static void main(String[] args) {
        Person person = new Person();

        person.name = "Arpit";
        person.age = 20;

        person.checkEligibility();
    }
}
```

Output-

Name: Arpit

Age: 20

Country: India

Voting Eligible: true

15) Create a class SimpleInterest with fields principal, rate, time, and a class variable bank. Use local variables to calculate interest.

```
class SimpleInterest {
    // Instance fields
    double principal;
    double rate;
    double time;

    // Class variable
    static String bank = "State Bank";

    // Constructor
    SimpleInterest(double principal, double rate, double time) {
        this.principal = principal;
        this.rate = rate;
        this.time = time;
    }

    void calculateInterest() {
        // Local variable
        double interest = (principal * rate * time) / 100;

        System.out.println("Bank: " + bank);
        System.out.println("Principal: " + principal);
        System.out.println("Rate: " + rate + "%");
        System.out.println("Time: " + time + " years");
        System.out.println("Simple Interest: " + interest);
    }

    public static void main(String[] args) {
        SimpleInterest obj = new SimpleInterest(10000, 5, 2);
        obj.calculateInterest();
    }
}
```

Output-

Bank: State Bank

Principal: 10000.0

Rate: 5.0%

Time: 2.0 years

Simple Interest: 1000.0

16) Create a class Triangle with sides a, b, c, and a class variable type = "Scalene/Isosceles/Equilateral". Use local variables to check validity.

```
class Triangle {
    // Instance fields
    int a, b, c;

    // Class variable
    static String type = "Scalene/Isosceles/Equilateral";

    Triangle(int a, int b, int c) {
        this.a = a;
        this.b = b;
        this.c = c;
    }

    void checkTriangle() {
        // Local variable for validity
        boolean valid = (a + b > c) && (b + c > a) && (a + c > b);

        if (!valid) {
            System.out.println("Invalid Triangle");
        }
        else if (a == b && b == c) {
            System.out.println("Triangle Type: Equilateral");
        }
        else if (a == b || b == c || a == c) {
            System.out.println("Triangle Type: Isosceles");
        }
        else {
            System.out.println("Triangle Type: Scalene");
        }
    }

    public static void main(String[] args) {
        Triangle obj = new Triangle(5, 5, 8);

        System.out.println("Sides: " + obj.a + ", " + obj.b + ", " + obj.c);
        obj.checkTriangle();
    }
}
```

Output-
Sides: 5, 5, 8

Triangle Type: Isosceles

17) Create a class StudentGrade with fields name, marks, and a class variable passingMarks. Use local variables to assign grade.

```
class StudentGrade {
    // Instance fields
    String name;
    int marks;

    // Class variable
    static int passingMarks = 40;

    StudentGrade(String name, int marks) {
        this.name = name;
        this.marks = marks;
    }

    void calculateGrade() {
        // Local variable
        String grade;

        if (marks < passingMarks) {
            grade = "F";
        } else if (marks >= 90) {
            grade = "A+";
        } else if (marks >= 80) {
            grade = "A";
        } else if (marks >= 70) {
            grade = "B";
        } else if (marks >= 60) {
            grade = "C";
        } else {
            grade = "D";
        }

        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
        System.out.println("Grade: " + grade);
    }

    public static void main(String[] args) {
        StudentGrade student = new StudentGrade("Arpit", 85);
        student.calculateGrade();
    }
}
```

```
}  
}
```

Output-

Name: Arpit

Marks: 85

Grade: A

18) Create a class ElectricBill with a field units and a class variable fixedCharge. Use local variables to calculate total bill.

```
class ElectricBill {  
    // Instance field  
    int units;  
  
    // Class variable  
    static double fixedCharge = 100;  
  
    ElectricBill(int units) {  
        this.units = units;  
    }  
  
    void calculateBill() {  
        // Local variables  
        double rate;  
        double energyCharge;  
        double totalBill;  
  
        if (units <= 100) {  
            rate = 3.0;  
        } else if (units <= 200) {  
            rate = 5.0;  
        } else {  
            rate = 7.0;  
        }  
  
        energyCharge = units * rate;  
        totalBill = energyCharge + fixedCharge;  
  
        System.out.println("Units Consumed: " + units);  
        System.out.println("Energy Charge: Rs. " + energyCharge);  
        System.out.println("Fixed Charge: Rs. " + fixedCharge);  
        System.out.println("Total Bill: Rs. " + totalBill);  
    }  
}
```

```

    }

    public static void main(String[] args) {
        ElectricBill bill = new ElectricBill(150);
        bill.calculateBill();
    }
}

```

output-

```

Units Consumed: 150
Energy Charge: Rs. 750.0
Fixed Charge: Rs. 100.0
Total Bill: Rs. 850.0

```

19) Create a class Product with fields id, name, price, and a class variable discountRate. Use local variables to calculate final price.

```

class Product {
    // Instance fields
    int id;
    String name;
    double price;

    // Class variable
    static double discountRate = 10.0; // 10% discount

    Product(int id, String name, double price) {
        this.id = id;
        this.name = name;
        this.price = price;
    }

    void calculateFinalPrice() {
        // Local variables
        double discount;
        double finalPrice;

        discount = (price * discountRate) / 100;
        finalPrice = price - discount;

        System.out.println("Product ID: " + id);
        System.out.println("Product Name: " + name);
        System.out.println("Original Price: Rs. " + price);
        System.out.println("Discount: Rs. " + discount);
        System.out.println("Final Price: Rs. " + finalPrice);
    }
}

```

```

    }

    public static void main(String[] args) {
        Product p = new Product(101, "Laptop", 50000);
        p.calculateFinalPrice();
    }
}

```

Output-

Product ID: 101

Product Name: Laptop

Original Price: Rs. 50000.0

Discount: Rs. 5000.0

Final Price: Rs. 45000.0

20) Create a class BankAccount with fields balance, and a class variable bankCode. Use local variables to ensure withdrawal does not exceed balance.

```

class BankAccount {
    // Instance field
    double balance;

    // Class variable
    static String bankCode = "SBI001";

    BankAccount(double balance) {
        this.balance = balance;
    }

    void withdraw(double amount) {
        // Local variable
        double remainingBalance = balance - amount;

        if (amount <= 0) {
            System.out.println("Invalid withdrawal amount.");
        } else if (amount > balance) {
            System.out.println("Withdrawal failed: Insufficient balance.");
        } else {
            balance = remainingBalance;

            System.out.println("Withdrawal successful.");
            System.out.println("Withdrawn Amount: Rs. " + amount);
            System.out.println("Remaining Balance: Rs. " + balance);
        }
    }
}

```

```

    }

    public static void main(String[] args) {
        BankAccount account = new BankAccount(10000);

        System.out.println("Bank Code: " + bankCode);
        System.out.println("Initial Balance: Rs. " + account.balance);

        account.withdraw(3000);
    }
}

```

Output-

```

Bank Code: SBI001
Initial Balance: Rs. 10000.0
Withdrawal successful.
Withdrawn Amount: Rs. 3000.0
Remaining Balance: Rs. 7000.0

```

21) Create a class Palindrome with a method using local variables to check if a number is palindrome. Add a class variable countChecks.

```

class Palindrome {
    // Class variable
    static int countChecks = 0;

    void checkPalindrome(int number) {
        // Local variables
        int original = number;
        int reverse = 0;
        int digit;

        countChecks++;

        while (number != 0) {
            digit = number % 10;
            reverse = reverse * 10 + digit;
            number = number / 10;
        }

        if (original == reverse) {
            System.out.println(original + " is a Palindrome.");
        } else {
            System.out.println(original + " is not a Palindrome.");
        }
    }
}

```

```

    }

    System.out.println("Number of checks: " + countChecks);
}

public static void main(String[] args) {
    Palindrome p = new Palindrome();

    p.checkPalindrome(121);
}
}

```

Output-

121 is a Palindrome.

Number of checks: 1

22) Create a class Armstrong with a method using local variables to check Armstrong number. Add a class variable totalChecks.

```

class Armstrong {
    // Class variable
    static int totalChecks = 0;

    void checkArmstrong(int number) {
        // Local variables
        int original = number;
        int temp = number;
        int sum = 0;
        int digit;
        int count = 0;

        totalChecks++;

        // Count number of digits
        while (temp != 0) {
            count++;
            temp = temp / 10;
        }

        temp = number;

        // Calculate Armstrong sum
        while (temp != 0) {

```

```

        digit = temp % 10;
        sum = sum + (int) Math.pow(digit, count);
        temp = temp / 10;
    }

    if (sum == original) {
        System.out.println(original + " is an Armstrong number.");
    } else {
        System.out.println(original + " is not an Armstrong number.");
    }

    System.out.println("Total checks: " + totalChecks);
}

public static void main(String[] args) {
    Armstrong a = new Armstrong();

    a.checkArmstrong(153);
}
}

```

Output-

153 is an Armstrong number.

Total checks: 1

23) Create a class PrimeChecker with a method using local variables to check prime. Add a class variable totalPrimeChecks.

```

class PrimeChecker {
    // Class variable
    static int totalPrimeChecks = 0;

    void checkPrime(int number) {
        // Local variables
        boolean isPrime = true;
        int i;

        totalPrimeChecks++;

        if (number <= 1) {
            isPrime = false;
        } else {
            for (i = 2; i <= number / 2; i++) {
                if (number % i == 0) {

```

```

        isPrime = false;
        break;
    }
}

if (isPrime) {
    System.out.println(number + " is a Prime number.");
} else {
    System.out.println(number + " is not a Prime number.");
}

System.out.println("Total Prime Checks: " + totalPrimeChecks);
}

public static void main(String[] args) {
    PrimeChecker p = new PrimeChecker();

    p.checkPrime(17);
}
}

```

Output-
 17 is a Prime number.
 Total Prime Checks: 1

24) Create a class Fibonacci with a method to display first N terms using local variables. Add a class variable seriesCount.

```

class Fibonacci {
    // Class variable
    static int seriesCount = 0;

    void displaySeries(int n) {
        // Local variables
        int first = 0;
        int second = 1;
        int next;
        int i;

        seriesCount++;

        System.out.println("Fibonacci Series:");
    }
}

```

```

        for (i = 1; i <= n; i++) {
            System.out.print(first + " ");

            next = first + second;
            first = second;
            second = next;
        }

        System.out.println();
        System.out.println("Series Count: " + seriesCount);
    }

    public static void main(String[] args) {
        Fibonacci f = new Fibonacci();

        f.displaySeries(10);
    }
}

```

Output-

Fibonacci Series:

0 1 1 2 3 5 8 13 21 34

Series Count: 1

25) Create a class Factorial with a method to compute factorial using local variables. Add a class variable totalCalls.

```

class Factorial {
    // Class variable
    static int totalCalls = 0;

    void calculateFactorial(int number) {
        // Local variables
        int factorial = 1;
        int i;

        totalCalls++;

        for (i = 1; i <= number; i++) {
            factorial = factorial * i;
        }

        System.out.println("Number: " + number);
    }
}

```

```

        System.out.println("Factorial: " + factorial);
        System.out.println("Total Calls: " + totalCalls);
    }

    public static void main(String[] args) {
        Factorial f = new Factorial();

        f.calculateFactorial(5);
    }
}

```

Output-

Number: 5

Factorial: 120

Total Calls: 1

26) Create a class Matrix with fields for 2x2 matrix and a class variable matrixType. Use local variables to add and subtract two matrices.

```

class Matrix {
    // Fields for 2x2 matrix
    int[][] matrix = new int[2][2];

    // Class variable
    static String matrixType = "2x2 Matrix";

    Matrix(int[][] values) {
        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                matrix[i][j] = values[i][j];
            }
        }
    }

    void addAndSubtract(Matrix m) {
        // Local variables
        int[][] sum = new int[2][2];
        int[][] difference = new int[2][2];

        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                sum[i][j] = matrix[i][j] + m.matrix[i][j];
                difference[i][j] = matrix[i][j] - m.matrix[i][j];
            }
        }
    }
}

```

```

    }

    System.out.println("Matrix Type: " + matrixType);

    System.out.println("Addition of Matrices:");
    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 2; j++) {
            System.out.print(sum[i][j] + " ");
        }
        System.out.println();
    }

    System.out.println("Subtraction of Matrices:");
    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 2; j++) {
            System.out.print(difference[i][j] + " ");
        }
        System.out.println();
    }
}

public static void main(String[] args) {
    int[][] a = {
        {1, 2},
        {3, 4}
    };

    int[][] b = {
        {5, 6},
        {7, 8}
    };

    Matrix m1 = new Matrix(a);
    Matrix m2 = new Matrix(b);

    m1.addAndSubtract(m2);
}
}

```

Output-

Matrix Type: 2x2 Matrix

Addition of Matrices:

6 8

10 12

Subtraction of Matrices:

-4 -4

-4 -4

27) Create a class StringReverse with a method using local variables to reverse a string. Add a class variable totalReversals.

```
class StringReverse {  
    // Class variable  
    static int totalReversals = 0;  
  
    void reverseString(String text) {  
        // Local variables  
        String reversed = "";  
        int i;  
  
        totalReversals++;  
  
        for (i = text.length() - 1; i >= 0; i--) {  
            reversed = reversed + text.charAt(i);  
        }  
  
        System.out.println("Original String: " + text);  
        System.out.println("Reversed String: " + reversed);  
        System.out.println("Total Reversals: " + totalReversals);  
    }  
  
    public static void main(String[] args) {  
        StringReverse s = new StringReverse();  
  
        s.reverseString("HELLO");  
    }  
}
```

Output-

Original String: HELLO

Reversed String: OLLEH

Total Reversals: 1

28) Create a class WordCounter with a method using local variables to count words in a sentence. Add a class variable language.

```
class WordCounter {
    // Class variable
    static String language = "English";

    void countWords(String sentence) {
        // Local variables
        String[] words = sentence.trim().split("\\s+");
        int count = words.length;

        System.out.println("Language: " + language);
        System.out.println("Sentence: " + sentence);
        System.out.println("Number of Words: " + count);
    }

    public static void main(String[] args) {
        WordCounter w = new WordCounter();

        w.countWords("Java is a simple programming language");
    }
}
```

Output-

Language: English

Sentence: Java is a simple programming language

Number of Words: 6

29) Create a class Employee with fields name, salary, and a class variable organization. Use local variables to compare salaries of two employees.

```
class Employee {
    // Instance fields
    String name;
    double salary;

    // Class variable
    static String organization = "ABC Company";

    Employee(String name, double salary) {
        this.name = name;
        this.salary = salary;
    }
}
```

```

    }

    void compareSalary(Employee e) {
        // Local variables
        double salary1 = this.salary;
        double salary2 = e.salary;

        System.out.println("Organization: " + organization);
        System.out.println(this.name + " Salary: Rs. " + salary1);
        System.out.println(e.name + " Salary: Rs. " + salary2);

        if (salary1 > salary2) {
            System.out.println(this.name + " has a higher salary.");
        } else if (salary2 > salary1) {
            System.out.println(e.name + " has a higher salary.");
        } else {
            System.out.println("Both employees have the same salary.");
        }
    }
}

public static void main(String[] args) {
    Employee emp1 = new Employee("Rahul", 50000);
    Employee emp2 = new Employee("Amit", 45000);

    emp1.compareSalary(emp2);
}
}

```

Output-

```

Organization: ABC Company
Rahul Salary: Rs. 50000.0
Amit Salary: Rs. 45000.0
Rahul has a higher salary.

```

30) Create a class Library with fields booksAvailable, and a class variable libraryName. Use local variables to update issue and return operations.

```

class Library {
    int booksAvailable;
    static String libraryName = "City Library";

    Library(int booksAvailable) {
        this.booksAvailable = booksAvailable;
    }
}

```

```

    }

    void issueBook() {
        int booksToIssue = 1; // local variable

        if (booksAvailable >= booksToIssue) {
            booksAvailable -= booksToIssue;
            System.out.println("Book issued successfully.");
        } else {
            System.out.println("No books available.");
        }
    }

    void returnBook() {
        int booksReturned = 1; // local variable
        booksAvailable += booksReturned;
        System.out.println("Book returned successfully.");
    }

    void display() {
        System.out.println("Library: " + libraryName);
        System.out.println("Books Available: " + booksAvailable);
    }

    public static void main(String[] args) {
        Library lib = new Library(5);

        lib.display();
        lib.issueBook();
        lib.display();
        lib.returnBook();
        lib.display();
    }
}

```

Output-

```

Library: City Library
Books Available: 5
Book issued successfully.
Library: City Library
Books Available: 4
Book returned successfully.
Library: City Library

```

Books Available: 5